① $R(s) = \frac{10}{s+2}$, $H(s) = \frac{3}{s+12}$

a) $Y(s) = \left(\frac{1}{s}\right)(H(s))(R(s)) = \left(\frac{1}{s}\right)\left(\frac{3}{s+12}\right)\left(\frac{10}{s+2}\right) = \boxed{\frac{30}{s(s+12)(s+2)}}$

b) The poles are $s = 0, -12, -2$

$\frac{30}{0(0+12)(0+2)} = \frac{30}{0}$ $\frac{30}{-12(-12+12)(-12+2)} = \frac{30}{0}$ $\frac{30}{-2(-2+12)(-2+2)} = \frac{30}{0}$

c) $\frac{30}{s(s+12)(s+2)} = \frac{K_1}{s} + \frac{K_2}{s+12} + \frac{K_3}{s+2}$

$K_1 = s\left[\frac{30}{s(s+12)(s+2)}\right]_{s=0} = \frac{30}{12 \cdot 2} = \frac{30}{24} = 15/12$

$K_2 = (s+12)\left[\frac{30}{s(s+12)(s+2)}\right]_{s=-12} = \frac{30}{-12(-10)} = \frac{30}{120} = 1/4$

$K_3 = (s+2)\left[\frac{30}{s(s+12)(s+2)}\right]_{s=-2} = \frac{30}{-2(-2+12)} = \frac{30}{-20} = -3/2$

$Y(s) = \frac{15}{12s} + \frac{1}{4(s+12)} + \frac{-3}{2(s+2)}$

$\mathcal{L}^{-1}$ → $\boxed{y(t) = \frac{15}{12} + \frac{e^{-12t}}{4} + \frac{-3e^{-2t}}{2}}$

d) $\lim_{s \to 0} sy(s) = \left[\frac{15s}{12s} + \frac{s}{4(s+12)} + \frac{-3s}{2(s+12)}\right]_{s=0}$

$= \frac{15}{12} + 0 + 0 = \boxed{15/12}$

$\lim_{t \to \infty} y(t) = \frac{15}{12} + 0 + 0 = \frac{15}{12}$ FVT matches $y(\infty)$